

YORK RIVER EAST RANGE LIGHT, VA, USA

WMO: 997708

Lat: 37.251N

Lon: 76.342W

Elev: 33

StdP: 14.68

Time Zone: -5.00 (NAE)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 21.9	(c) 25.6	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 35.0	(k) 33.8	(l) 32.6	(m) 34.6	(n) 16.8	(o) 0	(p) 0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 8.1	(c) 88.9	(d) N/A	(e) 86.8	(f) N/A	(g) 85.1	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 11.9	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 32.8	(b) 28.3	(c) 25.5	(e) 18.4	(f) 93.5	(g) 4.9	(h) 2.4	(i) 14.9	(j) 95.1	(k) 12.1	(l) 96.5	(m) 9.3	(n) 97.8	(o) 5.8	(p) 99.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.5	39.7	41.6	47.3	58.4	67.8	76.1	79.6	78.6	73.4	63.7	52.2	46.6
	DBStd	15.54	8.74	8.05	8.07	7.27	6.07	4.67	3.52	3.21	4.44	6.60	6.57	7.71
	HDD50	969	337	257	151	12	0	0	0	0	0	3	52	157
	HDD65	3312	786	657	549	221	42	0	0	0	1	100	385	571
	CDD50	4806	16	20	68	265	551	783	919	885	701	428	118	52
	CDD65	1676	0	0	1	24	128	334	454	420	253	61	1	0
	CDH74	11823	0	0	3	64	425	2306	4219	3502	1203	101	0	0
	CDH80	2525	0	0	0	5	61	499	1116	707	135	2	0	0
Wind	WSAvg	13.5	14.0	14.5	14.8	14.5	13.5	12.2	11.7	12.0	13.0	14.1	14.1	13.2
Precipitation	PrecAvg	46.3	3.3	3.0	3.9	3.1	3.5	3.9	5.4	5.1	5.0	3.7	2.9	3.5
	PrecMax	65.2	6.1	9.7	9.4	5.6	5.6	8.9	9.2	11.3	16.9	8.9	9.9	7.5
	PrecMin	31.4	1.4	1.2	0.7	0.7	1.1	1.4	1.1	1.2	0.9	0.1	0.1	1.4
	PrecStd	8.0	1.3	1.8	1.7	1.2	1.1	1.7	2.2	2.8	3.5	2.1	2.1	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.5	67.0	71.8	79.5	85.0	91.2	93.0	90.3	86.4	79.7	69.2	68.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	60.9	61.8	66.2	75.5	81.6	87.2	89.3	87.8	83.5	76.7	65.5	64.7
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	56.6	58.2	62.9	72.2	78.5	85.1	87.2	85.6	81.4	74.6	63.4	61.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	51.9	53.2	59.1	68.8	76.0	82.8	85.4	83.8	79.6	72.5	61.2	57.8
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	10.4	10.8	10.5	10.7	9.1	8.4	8.1	7.2	6.7	8.4	8.8	9.7
		MCDBR	16.8	17.2	16.9	15.2	12.3	11.7	11.3	10.1	9.4	10.2	10.9	14.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub		0.306	0.317	0.344	0.393	0.435	0.484	0.515	0.510	0.411	0.353	0.322	0.310
	taud		2.546	2.510	2.466	2.339	2.253	2.165	2.093	2.093	2.359	2.523	2.559	2.568
	Ebn at Noon		282	292	292	282	270	256	247	245	266	275	274	273
	Edh at Noon		24	29	33	39	43	48	51	49	36	28	24	23
All-Sky Solar Radiation	RadAvg		733	975	1304	1656	1872	2021	1928	1740	1412	1113	830	652
	RadStd		59	105	105	119	169	136	93	93	121	126	69	52

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (69 neighbors)		+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+284	+185

Nomenclature: See separate page